

The cube maximizes the Brownian exit-time/eigenvalue functional over all rectangles

Candidate solution of the rectangular subcase in arXiv:2003.06867

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Status. Candidate full proof of the rectangular subcase of Conjecture 5.4; substantial partial result for the full convex-domain conjecture. Likely valid, novelty uncertain; send to human review.

1 Source problem

Let B be standard Brownian motion in $\mathbb{R}^d$, let τ_D be its first exit time from D , and let $\lambda_1(D)$ be the first Dirichlet eigenvalue of $-\Delta$. For $a_k > 0$, put

$$R_{\mathbf{a}} = \prod_{k=1}^d (-a_k, a_k), \quad Q_d = (-1, 1)^d.$$

Remark 5.5 and equation (5.22) of Bañuelos–Mariano–Wang ask whether, for all $p > 0$,

$$\lambda_1(R_{\mathbf{a}})^p \mathbb{E}_0[\tau_{R_{\mathbf{a}}}^p] \leq \lambda_1(Q_d)^p \mathbb{E}_0[\tau_{Q_d}^p]. \tag{1}$$

The source says this was not verified even for $d = 2, p = 1$.

$$Q_d = \{(x_1, x_2, \dots, x_d) \in \mathbb{R}^d : |x_i| < 1\},$$

denotes the unit cube in $\mathbb{R}^d$.

5.5. Remarks on conjectures; rectangles, triangles, and ellipses.

Remark 5.5 (Rectangles). Conjecture 5.4 in general seems to be nontrivial. In fact, even the simplest case of rectangles does not seem obvious. More precisely, let $\mathbf{a} = (a_1, a_2, \dots, a_d)$, where $a_k > 0$ for all k . Set $R_{\mathbf{a}} = \{x = (x_1, x_2, \dots, x_d) : |x_k| < a_k, k = 1, \dots, d\}$. (We call $R_{\mathbf{a}}$ a rectangle.) Denote the origin in $\mathbb{R}^d$ by $\mathbf{0}$. In this case we would want to show that for all $\mathbf{a} \in \mathbb{R}^d$,

$$(5.22) \quad \lambda_1^p(R_{\mathbf{a}}) \mathbb{E}_{\mathbf{0}} [\tau_{R_{\mathbf{a}}}^p] \leq \lambda_1^p(Q_d) \mathbb{E}_{\mathbf{0}} [\tau_{Q_d}^p]$$

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with equality only when $R_{\mathbf{a}} = Q_d$. Since the eigenvalues of both $R_{\mathbf{a}}$ and Q_d are explicit and the components of the Brownian motion are independent, the inequality (5.22) can be stated in several different forms. Here is one. Let $I_{a_k} = (-a_k, a_k)$ and recall that $I = (-1, 1)$. Then (5.22) is equivalent to

$$(5.23) \quad \left(\sum_{k=1}^d \frac{1}{a_k^2} \right)^p \int_0^\infty p t^{p-1} \prod_{k=1}^d \mathbb{P}_0(\tau_{I_{a_k}} > t) dt \leq d^p \int_0^\infty p t^{p-1} (\mathbb{P}_0(\tau_I > t))^d dt.$$

Using the fact that $\mathbb{P}_0(\tau_{I_{a_k}} > t) = \mathbb{P}_0(\tau_I > \frac{t}{a_k})$ we may even assume that

$$a_1 = 1 < a_2 < \dots < a_d.$$

Using the fact that we know the heat kernel for an interval in terms of the eigenfunctions expansion (all which are explicitly given), the inequality has a rather appealing form. Let us look at the case $d = 2$ and $p = 1$. Then (5.23) is equivalent to

$$(5.24) \quad \begin{aligned} & (1 + a^2) \left[1 - \frac{32}{\pi^3} \sum_{n=0}^\infty \frac{(-1)^n}{(2n+1)^3} \operatorname{sech} \left[\left(n + \frac{1}{2} \right) \frac{\pi}{a} \right] \right], \\ & \leq 2 \left[1 - \frac{32}{\pi^3} \sum_{n=0}^\infty \frac{(-1)^n}{(2n+1)^3} \operatorname{sech} \left[\left(n + \frac{1}{2} \right) \pi \right] \right], \end{aligned}$$

for all $a > 1$.

Unfortunately, despite its simplicity and all its possible formulations, we have not been able to fully verify (5.22) for all rectangles even in the case $d = 2$ and $p = 1$.

Because the functional is invariant under common dilation, the unnormalized equality class must consist of all homothetic cubes. Under the source's later normalization $a_1 = 1$, this reduces to Q_d .

2 Idea of the proof

The rectangle problem tensorizes. If T is the exit time from $(-1, 1)$ and $S(t) = \mathbb{P}(T > t)$, then Brownian scaling and coordinate independence give

$$\mathbb{P}_0(\tau_{R_{\mathbf{a}}} > t) = \prod_{k=1}^d S(a_k^{-2}t).$$

Thus the desired balancing inequality follows immediately if $\log S$ is concave. The key observation is that T is an infinite convolution of exponential laws: its Laplace transform is $\operatorname{sech} \sqrt{2}s$, and Euler's product for cosh factors this into exponential Laplace transforms. Log-concavity survives convolution and weak limits, and the tail of a log-concave law is log-concave. Jensen then gives not merely all moment inequalities but a scale-free stochastic domination.

3 The interval exit time is log-concave

Lemma 1. *Let $T = \inf\{t > 0 : |B_t| \geq 1\}$ for one-dimensional standard Brownian motion with $B_0 = 0$. The law of T is log-concave on $[0, \infty)$, and its survival function $S(t) = \mathbb{P}(T > t)$ is log-concave on $[0, \infty)$.*

Proof. For $s \geq 0$, the function $u_s(x) = \mathbb{E}_x e^{-sT}$ solves

$$\frac{1}{2}u_s'' = su_s \quad (-1 < x < 1), \quad u_s(-1) = u_s(1) = 1.$$

Consequently

$$\mathbb{E}_0 e^{-sT} = u_s(0) = \frac{1}{\cosh \sqrt{2s}}.$$

Euler's product

$$\cosh z = \prod_{n=0}^{\infty} \left(1 + \frac{4z^2}{(2n+1)^2\pi^2}\right)$$

therefore gives

$$\mathbb{E}_0 e^{-sT} = \prod_{n=0}^{\infty} \frac{\lambda_n}{\lambda_n + s}, \quad \lambda_n = \frac{(2n+1)^2\pi^2}{8}. \quad (2)$$

Let X_n be independent exponential variables of rates λ_n . Since

$$\sum_{n \geq 0} \mathbb{E} X_n = \frac{8}{\pi^2} \sum_{n \geq 0} \frac{1}{(2n+1)^2} = 1,$$

$\sum_n X_n$ is finite almost surely. Equation (2) and uniqueness of Laplace transforms show

$$T \stackrel{d}{=} \sum_{n=0}^{\infty} X_n.$$

Every exponential density is log-concave. The Prékopa–Leindler theorem shows that convolution preserves log-concavity; log-concave probability measures are closed under weak convergence. Hence the law μ of T is log-concave.

Finally, for $x, y \geq 0$ and $0 < \theta < 1$, apply log-concavity of μ to the upper rays $A = (x, \infty)$ and $C = (y, \infty)$. Since

$$\theta A + (1 - \theta)C = (\theta x + (1 - \theta)y, \infty),$$

we obtain

$$S(\theta x + (1 - \theta)y) \geq S(x)^\theta S(y)^{1-\theta}.$$

Thus S is log-concave. □

4 Stochastic domination and the rectangle theorem

Theorem 1. *For every $d \geq 2$ and every axis-aligned rectangle $R_{\mathbf{a}}$,*

$$\lambda_1(R_{\mathbf{a}})\tau_{R_{\mathbf{a}}} \preceq_{\text{st}} \lambda_1(Q_d)\tau_{Q_d} \quad \text{under } \mathbb{P}_0. \quad (3)$$

Consequently, (1) holds for every $p > 0$. Equality in any one of these moment inequalities holds if and only if $a_1 = \dots = a_d$, that is, $R_{\mathbf{a}}$ is a homothetic cube.

Proof. Put $b_k = a_k^{-2}$ and $\bar{b} = d^{-1} \sum_k b_k$. The coordinate exit times are independent and Brownian scaling gives

$$\mathbb{P}_0(\tau_{R_{\mathbf{a}}} > t) = \prod_{k=1}^d S(b_k t).$$

By Lemma 1 and Jensen's inequality,

$$\prod_{k=1}^d S(b_k t) \leq S(\bar{b}t)^d = \mathbb{P}_0(\tau_{Q_d} > \bar{b}t). \quad (4)$$

Thus $\tau_{R_{\mathbf{a}}} \preceq_{\text{st}} \bar{b}^{-1} \tau_{Q_d}$. On the other hand, separation of variables gives

$$\lambda_1(R_{\mathbf{a}}) = \frac{\pi^2}{4} \sum_{k=1}^d b_k = \frac{\pi^2 d}{4} \bar{b}, \quad \lambda_1(Q_d) = \frac{\pi^2 d}{4}.$$

Multiplying the stochastic comparison by $\lambda_1(R_{\mathbf{a}})$ proves (3). Applying the increasing function $x \mapsto x^p$ proves (1) for every $p > 0$.

It remains to identify equality. If the b_k are all equal, every step is an equality. Conversely, suppose $m = \min b_k < M = \max b_k$ and equality holds for one moment. The tail-moment formula

$$\mathbb{E}X^p = p \int_0^\infty t^{p-1} \mathbb{P}(X > t) dt$$

and continuity force equality in (4) for all $t > 0$. Equality in Jensen for the concave function $\log S$ then makes $\log S$ affine on every interval $[mt, Mt]$. These intervals give overlapping neighborhoods of every point of $(0, \infty)$, so $\log S$ would be globally affine.

This is impossible. Indeed $T \geq X_0 + X_1$, whence

$$1 - S(t) = \mathbb{P}(T \leq t) \leq \mathbb{P}(X_0 \leq t, X_1 \leq t) = O(t^2),$$

so $(\log S)'(0+) = 0$, whereas $S(t) \rightarrow 0$ as $t \rightarrow \infty$. Therefore every moment inequality is strict unless all b_k , equivalently all a_k , are equal. $\square$

Corollary 1. *Among rectangles with a fixed first Dirichlet eigenvalue, the balanced cube maximizes the centered exit time in the usual stochastic order. Hence it maximizes $\mathbb{E}\Phi(\tau)$ for every increasing Φ for which the expectation exists, not only the positive moments.*

Maximizing starting point. For completeness, $x \mapsto \mathbb{P}_x(\tau_{(-a,a)} > t)$ is even and nonincreasing on $[0, a)$. This follows, for example, by applying the parabolic maximum principle to the spatial derivative of the killed heat solution. The rectangular survival probability is the product of the coordinate survival probabilities, so the origin maximizes it for every t , and hence maximizes every positive exit-time moment. The theorem therefore proves the source functional with its supremum over starting points as well.

5 Scope, verification, and novelty

The proof settles the full rectangular clause of Remark 5.5 for all $d \geq 2$ and all $p > 0$, and strengthens it to stochastic domination. It does not settle Conjecture 5.4 for arbitrary coordinate-symmetric convex domains. For such domains the independent-coordinate factorization is lost; moreover, domain inclusion or symmetrization moves λ_1 and the exit-time moment in opposite directions, so the rectangle argument does not formally extend.

An adversarial audit checked the Brownian/eigenvalue normalization, scaling, tail-moment identity, equality class, and the $0 < p < 1$ range. A bounded search through 11 August 2026 found the source and standard interval-exit formulas but no later proof of (5.22) or of the stochastic strengthening. Priority is not asserted. The accompanying numerical script checks constants and sample Jensen gaps but is not used in the proof.

References. R. Bañuelos, P. Mariano, and J. Wang, *Bounds for exit times of Brownian motion and the first Dirichlet eigenvalue for the Laplacian*, arXiv:2003.06867, especially Conjecture 5.4 and Remark 5.5.

A. Prékopa, *Logarithmic concave measures with applications to stochastic programming*, Acta Sci. Math. (Szeged) 32 (1971), 301–316.